PRAKTIKUM BASIS DATA

MODUL 3

(NAMA MODUL)

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PROGRAM STUDI S1 REKAYASA PERANGKAT LUNAK

DIREKTORAT KAMPUS PURWOKERTO

UNIVERSITAS TELKOM

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JURNAL/TP

1. RECORD

```
SQL> CREATE TABLE mahasiswa (
  2      id INT PRIMARY KEY,
  3      nama VARCHAR(50),
  4      tempat_lahir VARCHAR(50),
  5      tanggal_lahir DATE,
  6      nomor_hp VARCHAR(15),
  7      email VARCHAR(100),
  8      tinggi_badan INT,
  9      berat_badan INT
 10  );

     Table created.

     SQL> INSERT INTO mahasiswa VALUES
(1,'Andi','Jakarta',TO_DATE('2002-01-10', 'YYYY-MM-
DD'),'081234111','andi@gmail.com',170,60);

     1 row created.

     SQL> INSERT INTO mahasiswa VALUES
(2,'Budi','Bandung',TO_DATE('2001-02-11','YYYY-MM-
DD'),'081234112','budi@gmail.com',168,65);

     1 row created.

     SQL> INSERT INTO mahasiswa VALUES
(3,'Citra','Surabaya',TO_DATE('2002-03-12','YYYY-MM-
DD'),'081234113','citra@gmail.com',160,50);

     1 row created.

     SQL> INSERT INTO mahasiswa VALUES
(4,'Dina','Malang',TO_DATE('2001-04-13','YYYY-MM-
DD'),'081234114','dina@gmail.com',158,48);

     1 row created.

     SQL> INSERT INTO mahasiswa VALUES
(5,'Eko','Solo',TO_DATE('2002-05-14','YYYY-MM-
DD'),'081234115','eko@gmail.com',172,70);

     1 row created.

     SQL>
     SQL> INSERT INTO mahasiswa VALUES
(6,'Fajar','Yogyakarta',TO_DATE('2001-06-15','YYYY-MM-
DD'),'081234116','fajar@gmail.com',175,72);
```

```
38.      1 row created.
39.
40.      SQL> INSERT INTO mahasiswa VALUES
   (7,'Gina','Semarang',TO_DATE('2002-07-16','YYYY-MM-
   DD'),'081234117','gina@gmail.com',162,55);
41.
42.      1 row created.
43.
44.      SQL>
45.      SQL> INSERT INTO mahasiswa VALUES
   (8,'Hadi','Madiun',TO_DATE('2001-08-17','YYYY-MM-
   DD'),'081234118','hadi@gmail.com',169,66);
46.
47.      1 row created.
48.
49.      SQL> INSERT INTO mahasiswa VALUES
   (9,'Indah','Kediri',TO_DATE('2002-09-18','YYYY-MM-
   DD'),'081234119','indah@gmail.com',159,52);
50.
51.      1 row created.
52.
53.      SQL> INSERT INTO mahasiswa VALUES
   (10,'Joko','Blitar',TO_DATE('2001-10-19','YYYY-MM-
   DD'),'081234120','joko@gmail.com',173,68);
54.
55.      1 row created.
56.
57.      SQL> INSERT INTO mahasiswa VALUES
   (11,'Kiki','Bogor',TO_DATE('2002-11-20','YYYY-MM-
   DD'),'081234121','kiki@gmail.com',161,54);
58.
59.      1 row created.
60.
61.      SQL> INSERT INTO mahasiswa VALUES
   (12,'Lina','Depok',TO_DATE('2001-12-21','YYYY-MM-
   DD'),'081234122','lina@gmail.com',157,47);
62.
63.      1 row created.
64.
65.      SQL> INSERT INTO mahasiswa VALUES
   (13,'Mira','Bekasi',TO_DATE('2002-01-22','YYYY-MM-
   DD'),'081234123','mira@gmail.com',163,53);
66.
67.      1 row created.
68.
69.      SQL> INSERT INTO mahasiswa VALUES
   (14,'Nanda','Tegal',TO_DATE('2001-02-23','YYYY-MM-
   DD'),'081234124','nanda@gmail.com',171,69);
70.
71.      1 row created.
72.
```

```
73.     SQL> INSERT INTO mahasiswa VALUES
   (15,'Oki','Cirebon',TO_DATE('2002-03-24','YYYY-MM-
   DD'),'081234125','oki@gmail.com',174,71);
74.
75.     1 row created.
76.
77.     SQL> SELECT LOWER(nama) FROM mahasiswa;
78.
79.     LOWER(NAMA)
80.     --------------------------------------------------
81.     andi
82.     budi
83.     citra
84.     dina
85.     eko
86.     fajar
87.     gina
88.     hadi
89.     indah
90.     joko
91.     kiki
92.
93.     LOWER(NAMA)
94.     --------------------------------------------------
95.     lina
96.     mira
97.     nanda
98.     oki
99.
100.    15 rows selected.
101.
102.    SQL> SELECT UPPER(nama) FROM mahasiswa;
103.
104.    UPPER(NAMA)
105.    --------------------------------------------------
106.    ANDI
107.    BUDI
108.    CITRA
109.    DINA
110.    EKO
111.    FAJAR
112.    GINA
113.    HADI
114.    INDAH
115.    JOKO
116.    KIKI
117.
118.    UPPER(NAMA)
119.    --------------------------------------------------
120.    LINA
121.    MIRA
122.    NANDA
```

```
OKI

15 rows selected.

SQL> SELECT INITCAP(nama) FROM mahasiswa;

INITCAP(NAMA)
--------------------------------------------------
Andi
Budi
Citra
Dina
Eko
Fajar
Gina
Hadi
Indah
Joko
Kiki

INITCAP(NAMA)
--------------------------------------------------
Lina
Mira
Nanda
Oki

15 rows selected.

SQL> SELECT CONCAT(nama, email) FROM mahasiswa;

CONCAT(NAMA,EMAIL)
------------------------------------------------------------
----------------------
Andiandi@gmail.com
Budibudi@gmail.com
Citracitra@gmail.com
Dinadina@gmail.com
Ekoeko@gmail.com
Fajarfajar@gmail.com
Ginagina@gmail.com
Hadihadi@gmail.com
Indahindah@gmail.com
Jokojoko@gmail.com
Kikikiki@gmail.com

CONCAT(NAMA,EMAIL)
------------------------------------------------------------
----------------------
Linalina@gmail.com
Miramira@gmail.com
Nandananda@gmail.com
```

```
173.     Okioki@gmail.com
174.
175.     15 rows selected.
176.
177.     SQL> SELECT nama, LENGTH(nama) FROM mahasiswa;
178.
179.     NAMA
   LENGTH(NAMA)
180.     -------------------------------------------------- -------
   -----
181.     Andi
   4
182.     Budi
   4
183.     Citra
   5
184.     Dina
   4
185.     Eko
   3
186.     Fajar
   5
187.     Gina
   4
188.     Hadi
   4
189.     Indah
   5
190.     Joko
   4
191.     Kiki
   4
192.
193.     NAMA
   LENGTH(NAMA)
194.     -------------------------------------------------- -------
   -----
195.     Lina
   4
196.     Mira
   4
197.     Nanda
   5
198.     Oki
   3
199.
200.     15 rows selected.
201.
202.     SQL> SELECT SUBSTR(nama,1,3) FROM mahasiswa;
203.
204.     SUBSTR(NAMA,
205.     ------------
```

```
206.     And
207.     Bud
208.     Cit
209.     Din
210.     Eko
211.     Faj
212.     Gin
213.     Had
214.     Ind
215.     Jok
216.     Kik
217.
218.     SUBSTR(NAMA,
219.     ------------
220.     Lin
221.     Mir
222.     Nan
223.     Oki
224.
225.     15 rows selected.
226.
227.     SQL> SELECT LPAD(nama,10,'*') FROM mahasiswa;
228.
229.     LPAD(NAMA,10,'*')
230.     ----------------------------------------
231.     ******Andi
232.     ******Budi
233.     *****Citra
234.     ******Dina
235.     *******Eko
236.     *****Fajar
237.     ******Gina
238.     ******Hadi
239.     *****Indah
240.     ******Joko
241.     ******Kiki
242.
243.     LPAD(NAMA,10,'*')
244.     ----------------------------------------
245.     ******Lina
246.     ******Mira
247.     *****Nanda
248.     *******Oki
249.
250.     15 rows selected.
251.
252.     SQL> SELECT RPAD(nama,10,'*') FROM mahasiswa;
253.
254.     RPAD(NAMA,10,'*')
255.     ----------------------------------------
256.     Andi******
257.     Budi******
```

```
Citra*****
Dina******
Eko*******
Fajar*****
Gina******
Hadi******
Indah*****
Joko******
Kiki******

RPAD(NAMA,10,'*')
----------------------------------------
Lina******
Mira******
Nanda*****
Oki*******

15 rows selected.

SQL> SELECT LTRIM('   Naufal') FROM dual;

LTRIM(
------
Naufal

SQL> SELECT RTRIM('Naufal   ') FROM dual;

RTRIM(
------
Naufal

SQL> SELECT INSTR(email,'@') FROM mahasiswa;

INSTR(EMAIL,'@')
----------------
                5
                5
                6
                5
                4
                6
                5
                5
                6
                5
                5

INSTR(EMAIL,'@')
----------------
                5
                5
                6
```

```
                  4

15 rows selected.

SQL> SELECT REPLACE(email,'gmail','yahoo') FROM mahasiswa;

REPLACE(EMAIL,'GMAIL','YAHOO')
------------------------------------------------------------
----------------------
andi@yahoo.com
budi@yahoo.com
citra@yahoo.com
dina@yahoo.com
eko@yahoo.com
fajar@yahoo.com
gina@yahoo.com
hadi@yahoo.com
indah@yahoo.com
joko@yahoo.com
kiki@yahoo.com

REPLACE(EMAIL,'GMAIL','YAHOO')
------------------------------------------------------------
----------------------
lina@yahoo.com
mira@yahoo.com
nanda@yahoo.com
oki@yahoo.com

15 rows selected.

SQL> SELECT TO_CHAR(tanggal_lahir,'MM') FROM mahasiswa;

TO
--
01
02
03
04
05
06
07
08
09
10
11

TO
--
12
01
02
```

```
03

15 rows selected.

SQL> SELECT TO_CHAR(tanggal_lahir,'DD') FROM mahasiswa;

TO
--
10
11
12
13
14
15
16
17
18
19
20

TO
--
21
22
23
24

15 rows selected.

SQL> SELECT TO_CHAR(tanggal_lahir,'YYYY') FROM mahasiswa;

TO_C
----
2002
2001
2002
2001
2002
2001
2002
2001
2002
2001
2002

TO_C
----
2001
2002
2001
2002
```

```
15 rows selected.

SQL> SELECT TO_NUMBER('12345') FROM dual;

TO_NUMBER('12345')
------------------
             12345

SQL> SELECT TO_CHAR(tanggal_lahir,'DD-MM-YYYY') FROM mahasiswa;

TO_CHAR(TA
----------
10-01-2002
11-02-2001
12-03-2002
13-04-2001
14-05-2002
15-06-2001
16-07-2002
17-08-2001
18-09-2002
19-10-2001
20-11-2002

TO_CHAR(TA
----------
21-12-2001
22-01-2002
23-02-2001
24-03-2002

15 rows selected.

SQL> SELECT TO_CHAR(tinggi_badan) FROM mahasiswa;

TO_CHAR(TINGGI_BADAN)
----------------------------------------
170
168
160
158
172
175
162
169
159
173
161

TO_CHAR(TINGGI_BADAN)
----------------------------------------
```

```
157
163
171
174

15 rows selected.

SQL> SELECT TO_DATE('10-01-2002','DD-MM-YYYY') FROM dual;

TO_DATE('
---------
10-JAN-02

SQL> SELECT ABS(-10) FROM dual;

  ABS(-10)
----------
        10

SQL> SELECT MOD(10,3) FROM dual;

 MOD(10,3)
----------
         1

SQL> -- FLOOR
SQL> SELECT FLOOR(10.7) FROM dual;

FLOOR(10.7)
-----------
         10

SQL> -- FLOOR
SQL> SELECT FLOOR(10.7) FROM dual;

FLOOR(10.7)
-----------
         10

SQL> SELECT POWER(2,3) FROM dual;

POWER(2,3)
----------
         8

SQL> SELECT ROUND(10.567,2) FROM dual;

ROUND(10.567,2)
---------------
          10.57

SQL> SELECT CEIL(10.2) FROM dual;
```

```
515.
516.     CEIL(10.2)
517.     ----------
518.             11
519.
520.     SQL> SELECT * FROM mahasiswa;
521.
522.             ID NAMA
523.     ---------- ------------------------------------------------
   ---
524.     TEMPAT_LAHIR
   TANGGAL_L NOMOR_HP
525.     ---------------------------------------------------- -------
   -- ---------------
526.     EMAIL
527.     ------------------------------------------------------------
   ----------------------
528.     TINGGI_BADAN BERAT_BADAN
529.     ------------ -----------
530.              1 Andi
531.     Jakarta                                            10-JAN-
   02 081234111
532.     andi@gmail.com
533.              170          60
534.
535.
536.             ID NAMA
537.     ---------- ------------------------------------------------
   ---
538.     TEMPAT_LAHIR
   TANGGAL_L NOMOR_HP
539.     ---------------------------------------------------- -------
   -- ---------------
540.     EMAIL
541.     ------------------------------------------------------------
   ----------------------
542.     TINGGI_BADAN BERAT_BADAN
543.     ------------ -----------
544.              2 Budi
545.     Bandung                                            11-FEB-
   01 081234112
546.     budi@gmail.com
547.              168          65
548.
549.
550.             ID NAMA
551.     ---------- ------------------------------------------------
   ---
552.     TEMPAT_LAHIR
   TANGGAL_L NOMOR_HP
553.     ---------------------------------------------------- -------
   -- ---------------
```

```
EMAIL
------------------------------------------------------------
----------------------
TINGGI_BADAN BERAT_BADAN
------------ -----------
             3 Citra
Surabaya                                          12-MAR-
02 081234113
citra@gmail.com
             160          50


            ID NAME
---------- ----------------------------------------------
---
TEMPAT_LAHIR
TANGGAL_L NOMOR_HP
-------------------------------------------------- -------
-- ---------------
EMAIL
------------------------------------------------------------
----------------------
TINGGI_BADAN BERAT_BADAN
------------ -----------
             4 Dina
Malang                                            13-APR-
01 081234114
dina@gmail.com
             158          48


            ID NAME
---------- ----------------------------------------------
---
TEMPAT_LAHIR
TANGGAL_L NOMOR_HP
-------------------------------------------------- -------
-- ---------------
EMAIL
------------------------------------------------------------
----------------------
TINGGI_BADAN BERAT_BADAN
------------ -----------
             5 Eko
Solo                                              14-MAY-
02 081234115
eko@gmail.com
             172          70


            ID NAME
```

```
----------- ------------------------------------------------
---
TEMPAT_LAHIR
TANGGAL_L NOMOR_HP
------------------------------------------------- -------
-- ---------------
EMAIL
------------------------------------------------------------
----------------------
TINGGI_BADAN BERAT_BADAN
------------ -----------
             6 Fajar
Yogyakarta                                          15-JUN-
01 081234116
fajar@gmail.com
             175          72


            ID NAME
----------- ------------------------------------------------
---
TEMPAT_LAHIR
TANGGAL_L NOMOR_HP
------------------------------------------------- -------
-- ---------------
EMAIL
------------------------------------------------------------
----------------------
TINGGI_BADAN BERAT_BADAN
------------ -----------
             7 Gina
Semarang                                            16-JUL-
02 081234117
gina@gmail.com
             162          55


            ID NAME
----------- ------------------------------------------------
---
TEMPAT_LAHIR
TANGGAL_L NOMOR_HP
------------------------------------------------- -------
-- ---------------
EMAIL
------------------------------------------------------------
----------------------
TINGGI_BADAN BERAT_BADAN
------------ -----------
             8 Hadi
Madiun                                              17-AUG-
01 081234118
```

```
630.     hadi@gmail.com
631.              169          66
632.
633.
634.             ID NAME
635.     ---------- ------------------------------------------------
   ---
636.     TEMPAT_LAHIR
   TANGGAL_L NOMOR_HP
637.     -------------------------------------------------- -------
   -- ---------------
638.     EMAIL
639.     ------------------------------------------------------------
   ----------------------
640.     TINGGI_BADAN BERAT_BADAN
641.     ------------ -----------
642.              9 Indah
643.     Kediri                                                18-SEP-
   02 081234119
644.     indah@gmail.com
645.              159          52
646.
647.
648.             ID NAME
649.     ---------- ------------------------------------------------
   ---
650.     TEMPAT_LAHIR
   TANGGAL_L NOMOR_HP
651.     -------------------------------------------------- -------
   -- ---------------
652.     EMAIL
653.     ------------------------------------------------------------
   ----------------------
654.     TINGGI_BADAN BERAT_BADAN
655.     ------------ -----------
656.             10 Joko
657.     Blitar                                                19-OCT-
   01 081234120
658.     joko@gmail.com
659.              173          68
660.
661.
662.             ID NAME
663.     ---------- ------------------------------------------------
   ---
664.     TEMPAT_LAHIR
   TANGGAL_L NOMOR_HP
665.     -------------------------------------------------- -------
   -- ---------------
666.     EMAIL
667.     ------------------------------------------------------------
   ----------------------
```

```
668.     TINGGI_BADAN BERAT_BADAN
669.     ------------ -----------
670.             11 Kiki
671.     Bogor                                          20-NOV-
   02 081234121
672.     kiki@gmail.com
673.              161          54
674.
675.
676.             ID NAMA
677.     ---------- ------------------------------------------------
   ---
678.     TEMPAT_LAHIR
   TANGGAL_L NOMOR_HP
679.     -------------------------------------------------- -------
   -- ---------------
680.     EMAIL
681.     ------------------------------------------------------------
   ----------------------
682.     TINGGI_BADAN BERAT_BADAN
683.     ------------ -----------
684.             12 Lina
685.     Depok                                          21-DEC-
   01 081234122
686.     lina@gmail.com
687.              157          47
688.
689.
690.             ID NAMA
691.     ---------- ------------------------------------------------
   ---
692.     TEMPAT_LAHIR
   TANGGAL_L NOMOR_HP
693.     -------------------------------------------------- -------
   -- ---------------
694.     EMAIL
695.     ------------------------------------------------------------
   ----------------------
696.     TINGGI_BADAN BERAT_BADAN
697.     ------------ -----------
698.             13 Mira
699.     Bekasi                                         22-JAN-
   02 081234123
700.     mira@gmail.com
701.              163          53
702.
703.
704.             ID NAMA
705.     ---------- ------------------------------------------------
   ---
706.     TEMPAT_LAHIR
   TANGGAL_L NOMOR_HP
```

```
707.     ------------------------------------------------- -------
   -- ---------------
708.     EMAIL
709.     ------------------------------------------------------------
   ----------------------
710.     TINGGI_BADAN BERAT_BADAN
711.     ------------ -----------
712.             14 Nanda
713.     Tegal                                              23-FEB-
   01 081234124
714.     nanda@gmail.com
715.              171          69
716.
717.
718.             ID NAME
719.     ---------- ----------------------------------------------
   ---
720.     TEMPAT_LAHIR
   TANGGAL_L NOMOR_HP
721.     ------------------------------------------------- -------
   -- ---------------
722.     EMAIL
723.     ------------------------------------------------------------
   ----------------------
724.     TINGGI_BADAN BERAT_BADAN
725.     ------------ -----------
726.             15 Oki
727.     Cirebon                                            24-MAR-
   02 081234125
728.     oki@gmail.com
729.              174          71
730.
731.
732.     15 rows selected.
733.
734.     SQL> SET LINESIZE 200
735.     SQL> SET PAGESIZE 50
736.     SQL> COLUMN nama FORMAT A15
737.     SQL> COLUMN tempat_lahir FORMAT A15
738.     SQL> COLUMN nomor_hp FORMAT A15
739.     SQL> COLUMN email FORMAT A25
740.     SQL> SELECT * FROM mahasiswa;
741.
742.             ID NAME            TEMPAT_LAHIR    TANGGAL_L
   NOMOR_HP        EMAIL                     TINGGI_BADAN
   BERAT_BADAN
743.     ---------- --------------- --------------- --------- -----
   ---------- ------------------------- ------------ -----------
744.              1 Andi            Jakarta         10-JAN-02
   081234111       andi@gmail.com                     170
   60
```

```
745.              2 Budi            Bandung         11-FEB-01
   081234112       budi@gmail.com                     168
   65
746.              3 Citra           Surabaya        12-MAR-02
   081234113       citra@gmail.com                    160
   50
747.              4 Dina            Malang          13-APR-01
   081234114       dina@gmail.com                     158
   48
748.              5 Eko             Solo            14-MAY-02
   081234115       eko@gmail.com                      172
   70
749.              6 Fajar           Yogyakarta      15-JUN-01
   081234116       fajar@gmail.com                    175
   72
750.              7 Gina            Semarang        16-JUL-02
   081234117       gina@gmail.com                     162
   55
751.              8 Hadi            Madiun          17-AUG-01
   081234118       hadi@gmail.com                     169
   66
752.              9 Indah           Kediri          18-SEP-02
   081234119       indah@gmail.com                    159
   52
753.             10 Joko            Blitar          19-OCT-01
   081234120       joko@gmail.com                     173
   68
754.             11 Kiki            Bogor           20-NOV-02
   081234121       kiki@gmail.com                     161
   54
755.             12 Lina            Depok           21-DEC-01
   081234122       lina@gmail.com                     157
   47
756.             13 Mira            Bekasi          22-JAN-02
   081234123       mira@gmail.com                     163
   53
757.             14 Nanda           Tegal           23-FEB-01
   081234124       nanda@gmail.com                    171
   69
758.             15 Oki             Cirebon         24-MAR-02
   081234125       oki@gmail.com                      174
   71
759.
760.     15 rows selected.
761.
762.     SQL> SPOOL OF
```